# Position, distance, displacement

**antocciphysics**

UNIT 1 / DAY 2 / HONORS + SEI

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Distinguish a location, a total path length, and a change in position.

**SUCCESS CRITERIA - I can...**

1. Label the origin, start, turn and finish.
2. Calculate path distance and signed displacement separately.
3. Explain why position and displacement can differ.

**Core:** teacher selects from 1-9. **Stretch:** 10 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “position” and “distance” to compare your model or result. Sentence starters are optional.

**position**: location measured from a chosen origin

**distance**: total length of the path traveled

**displacement**: change from initial position to final position, with direction

## 01 Bring back an earlier idea

Positions are 15, 10, 5 m at 0, 2, 4 s.
**Predict** position at 7 s.
**Give** the rate, an assumption and a future test.

**Sentence starter:** I predict … . My model supports this because … .

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $\Delta x = x_f - x_i$; distance = sum of path lengths. Choose an origin and positive direction. i = initial; f = final.

**Worked reasoning:** A walker starts at 2 m, reaches 6 m, then ends at 4 m. Distance=4+2=6 m. Displacement=4-2=2 m right.

**Your turn: Draw** the two route arrows and a separate start-to-finish arrow.
**Label** the length of each.

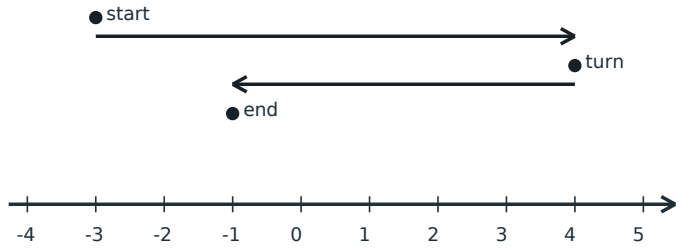
**Sentence starter:** The … in my model represents … . Its label shows … .

**Marzano check - circle:** ① ② ③

## 03 Track a signed route

East is positive. Start at -3 m, go to +4 m, then end at -1 m.
**Sketch** the route and calculate distance and displacement.

**Sentence starter:** I found … [unit]. This value means … in this situation.




Position (m); right is positive

**Marzano check - circle:** ① ② ③

## 04 Change the coordinate convention

For the route in 3, reverse the positive direction while keeping the origin fixed.
**Give** the new endpoint coordinates and displacement.
What stays unchanged?

**Sentence starter:** I notice … . This supports … because … .

**Marzano check - circle:** ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

# Position, distance, displacement

**antocciphysics**



Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 A closed path

A runner completes one 400 m lap, then runs another 100 m around a circular track. Is the displacement magnitude 500 m?
**Explain** what would be needed to calculate it.

**Word help:** magnitude = size of a quantity without its direction

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .

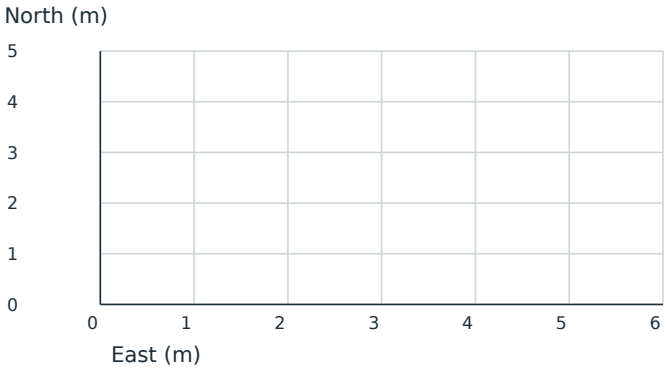
**Marzano check - circle:** 1 2 3

## 06 Different paths, same endpoints

**Sketch** two different routes from (0, 0) m to (4, 3) m.
**Give** each path distance and the common displacement magnitude.

**Word help:** magnitude = size of a quantity without its direction

**Sentence starter:** Both ... . They differ in ... because ... .




**Marzano check - circle:** 1 2 3

## 07 Zero and nonzero

Can displacement be zero while distance is nonzero? Can distance be zero while displacement is nonzero? Justify both answers.

**Word help:** justify = support your answer with a reason or evidence

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .

**Marzano check - circle:** 1 2 3

## 08 Choose a method

A fictional observing report records 3 images of a moving object but gives no distance or size calibration.
**State** a supported observation and explain why a precise impact prediction is unsupported.

**Word help:** prediction = what a model says may happen next

**Sentence starter:** I predict ... . My model supports this because ... .

**Marzano check - circle:** 1 2 3

## 09 Exit check

Start at +5 m, walk to -2 m, and finish at +1 m.
**Calculate** distance and displacement.
**Explain** why neither equals final position.

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .

**Marzano check - circle:** 1 2 3

## 10 STRETCH: Apply it another way

In one dimension, start at -2 m and end at +3 m with total distance 11 m.
**Find** two possible single-turn positions and draw both routes.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** 1 2 3